

NaijaCart Data Dictionary

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NaijaCart Data Dictionary

This document describes every table and field in the **NaijaCart** dataset, an e-commerce dataset covering orders, customers, products, stores, promotions, payments, employees, and delivery logistics across Nigeria. The dataset is structured as a star/snowflake schema with 3 fact tables and 8 dimension tables.

✦ Before loading these tables into the Power BI model, review `NaijaCart_Data_Cleaning_Log.md` for the transformation steps to apply in Power Query.

Table Overview

Table	Type	Rows	Grain
Orders	Fact	15,500	One row per order
OrderLines	Fact	26,239	One row per product per order
Delivery	Fact	13,000	One row per delivered/attempted order (excludes cancelled orders)
Customers	Dimension	3,500	One row per customer
Products	Dimension	405	One row per product
Stores	Dimension	30	One row per store/seller
Promotions	Dimension	12	One row per promotion campaign
PaymentMethods	Dimension	5	One row per payment method

Employees	Dimension	70	One row per staff member
Riders	Dimension	90	One row per delivery rider/agent
Date	Dimension	1,318	One row per calendar day (Jan 1, 2023 – Aug 10, 2026)

Orders (Fact)

One row per order placed. Grain: **1 row = 1 order**.

Column	Data Type	Description
OrderID	Text	Unique order identifier (e.g. ORD-000001). Primary key. Sequential and chronological — increases in step with OrderDate.
OrderDate	Date	Date the order was placed.
CustomerID	Text	Foreign key to Customers[CustomerID].
StoreID	Text	Foreign key to Stores[StoreID]. Store/seller that fulfilled the order.
EmployeeID	Text	Foreign key to Employees[EmployeeID]. Staff member who processed the order.
PromotionID	Text	Foreign key to Promotions[PromotionID]. NO_PROMO if no promotion applied.
PaymentMethodID	Text	Foreign key to PaymentMethods[PaymentMethodID].
OrderStatus	Text	One of: Completed, Cancelled, Returned.
ShippingFee	Decimal	Shipping cost charged on the order (¥). 0 for cancelled orders.
OrderTotal	Decimal	Sum of all LineTotal values across the order's items (¥). Excludes shipping fee.

OrderLines (Fact)

One row per product within an order. Grain: **1 row = 1 product line item on 1 order**.

Column	Data Type	Description
OrderLineID	Text	Unique line item identifier (e.g. OL-0000001). Primary key.
OrderID	Text	Foreign key to Orders[OrderID].
ProductID	Text	Foreign key to Products[ProductID].

Quantity	Whole Number	Units of the product purchased on this line.
UnitPrice	Decimal	Price per unit at time of sale (₦), copied from Products[UnitPrice] .
DiscountRate	Decimal	Discount rate applied to this line (0–1, e.g. 0.15 = 15% off), inherited from the order's promotion.
LineTotal	Decimal	$\text{Quantity} \times \text{UnitPrice} \times (1 - \text{DiscountRate})$ (₦).

Delivery (Fact)

One row per order that was dispatched (excludes Cancelled orders). Grain: **1 row = 1 order's delivery attempt**.

Column	Data Type	Description
DeliveryID	Text	Unique delivery record identifier (e.g. DEL-000001). Primary key.
OrderID	Text	Foreign key to Orders[OrderID] .
RiderID	Text	Foreign key to Riders[RiderID] .
DispatchDate	Date	Date the order left the store/warehouse.
DeliveryDate	Date	Date the order was delivered. Blank if DeliveryStatus is Delayed or Failed .
DestinationState	Text	Nigerian state the order was delivered to (matches the customer's state).
DestinationCity	Text	City the order was delivered to.
DistanceKM	Decimal	Distance in kilometres between store and delivery destination.
DeliveryTimeHours	Decimal	Hours elapsed between dispatch and delivery. Blank if not delivered.
DeliveryStatus	Text	One of: Delivered , Delayed , Failed .
DeliveryFee	Decimal	Fee charged for delivery (₦), equal to the order's ShippingFee .

Customers (Dimension)

Column	Data Type	Description
CustomerID	Text	Unique customer identifier

		(e.g. CUST-00001). Primary key.
CustomerName	Text	Full name of the customer.
Gender	Text	Male or Female .
State	Text	Nigerian state of residence.
City	Text	City of residence.
Region	Text	Geopolitical zone (South West, South East, South South, North Central, North West, North East).
SignupDate	Date	Date the customer registered on the platform.
CustomerSegment	Text	New , Returning , or VIP .

Products (Dimension)

💰 All monetary values in this table (and throughout the dataset) are in **Nigerian Naira (₦)**. When building measures or formatting visuals in Power BI, set the number format to Naira rather than the default currency symbol.

Column	Data Type	Description
ProductID	Text	Unique product identifier (e.g. PRD-00001). Primary key.
ProductName	Text	Product display name (brand + subcategory + variant).
Category	Text	Top-level category (e.g. Phones & Tablets , Fashion , Groceries). 9 categories total.
SubCategory	Text	Category subdivision (e.g. Smartphones , Men's Clothing). Price bands are set at this level — e.g. Snacks and Phone Accessories are priced far lower than Laptops or Furniture .
Brand	Text	Product brand.
CostPrice	Decimal	Cost to the business per unit (₦).
UnitPrice	Decimal	Selling price per unit before discount (₦).

Stores (Dimension)

Column	Data Type	Description
StoreID	Text	Unique store/seller identifier (e.g. STO-001). Primary key.
StoreName	Text	Store/seller display name.
SellerType	Text	Official Store or Third-Party Seller .
State	Text	Nigerian state the store operates

		from.
City	Text	City the store operates from.
Region	Text	Geopolitical zone.

Promotions (Dimension)

Column	Data Type	Description
PromotionID	Text	Unique promotion identifier (e.g. PROMO-00). Primary key. NO_PROMO represents no promotion.
PromotionName	Text	Campaign name (e.g. Black Friday , Dettý December Sale , Ramadan Sale).
DiscountType	Text	Percentage or None .
DiscountValue	Decimal	Discount rate as a fraction (e.g. 0.25 = 25% off). 0 for NO_PROMO .
StartDate	Date	Campaign start date. Blank for NO_PROMO .
EndDate	Date	Campaign end date. Blank for NO_PROMO .

PaymentMethods (Dimension)

Column	Data Type	Description
PaymentMethodID	Text	Unique payment method identifier (e.g. PM-01). Primary key.
PaymentMethodName	Text	Card Payment , Bank Transfer , USSD , Pay on Delivery , or Mobile Wallet .

Employees (Dimension)

Column	Data Type	Description
EmployeeID	Text	Unique employee identifier (e.g. EMP-0001). Primary key.
EmployeeName	Text	Full name of the employee.
Role	Text	Warehouse Staff , Customer Service Agent , Store Manager , or Fulfillment Associate .
StoreID	Text	Foreign key to Stores[StoreID] . Store the employee is assigned to.
State	Text	State of the assigned store.

HireDate	Date	Date the employee was hired.
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Riders (Dimension)

Column	Data Type	Description
RiderID	Text	Unique rider/delivery agent identifier (e.g. RDR-0001). Primary key.
RiderName	Text	Full name of the rider.
VehicleType	Text	Motorbike, Van, Tricycle (Keke), or Bicycle.
LogisticsPartner	Text	In-House Fleet, GIG Logistics, Kwik Delivery, Max.ng, or DHL Nigeria.
State	Text	Rider's base state.
City	Text	Rider's base city.

Date (Dimension)

Standard calendar date table spanning **January 1, 2023 – August 10, 2026**. (Extends 10 days past the last OrderDate so that every Delivery[DeliveryDate] — which can fall after the order date — still has a matching row for relationships.)

Column	Data Type	Description
DateKey	Whole Number	Integer key in YYYYMMDD format (e.g. 20230101). Recommended key for relationships.
Date	Date	Calendar date.
Day	Whole Number	Day of month (1–31).
Month	Whole Number	Month number (1–12).
MonthName	Text	Full month name (e.g. January).
Quarter	Whole Number	Calendar quarter (1–4).
Year	Whole Number	Calendar year.
Weekday	Text	Full weekday name (e.g. Monday).
IsWeekend	Boolean	TRUE if Saturday or Sunday, else FALSE.

Table Relationships (for Power BI Model View)

From Table	From Column	To Table	To Column	Cardinality
Orders	CustomerID	Customers	CustomerID	Many-to-One
Orders	StoreID	Stores	StoreID	Many-to-One

Orders	EmployeeID	Employees	EmployeeID	Many-to-One
Orders	PromotionID	Promotions	PromotionID	Many-to-One
Orders	PaymentMethodID	PaymentMethods	PaymentMethodID	Many-to-One
Orders	OrderDate	Date	Date	Many-to-One
OrderLines	OrderID	Orders	OrderID	Many-to-One
OrderLines	ProductID	Products	ProductID	Many-to-One
Delivery	OrderID	Orders	OrderID	One-to-One
Delivery	RiderID	Riders	RiderID	Many-to-One
Employees	StoreID	Stores	StoreID	Many-to-One

Recommended Power BI relationship direction: single-direction filtering, flowing from each dimension table into `Orders`, `OrderLines`, and `Delivery`. `Orders` acts as the central fact table connecting to most dimensions; `OrderLines` and `Delivery` connect to `Orders` and inherit its filter context.

Data Cleaning

Before loading these tables into the Power BI model, several data quality issues in the raw source must be assessed and cleaned in Power Query. See `NaijaCart_Data_Cleaning_Log.md` for the full data quality assessment and step-by-step transformation log.

Date Integrity Checks (Validated)

The following referential date-logic checks were run and confirmed clean:

- No `Orders[OrderDate]` precedes the ordering customer's `Customers[SignupDate]`.
- No `Orders[OrderDate]` precedes the processing employee's `Employees[HireDate]`.
- No `Delivery[DispatchDate]` precedes its order's `Orders[OrderDate]`.
- No `Delivery[DeliveryDate]` precedes its own `DispatchDate`.
- Every `Delivery[DeliveryDate]` falls within the `Date` dimension's range.
- `Orders[OrderID]` is sequential and chronologically consistent with `Orders[OrderDate]`.